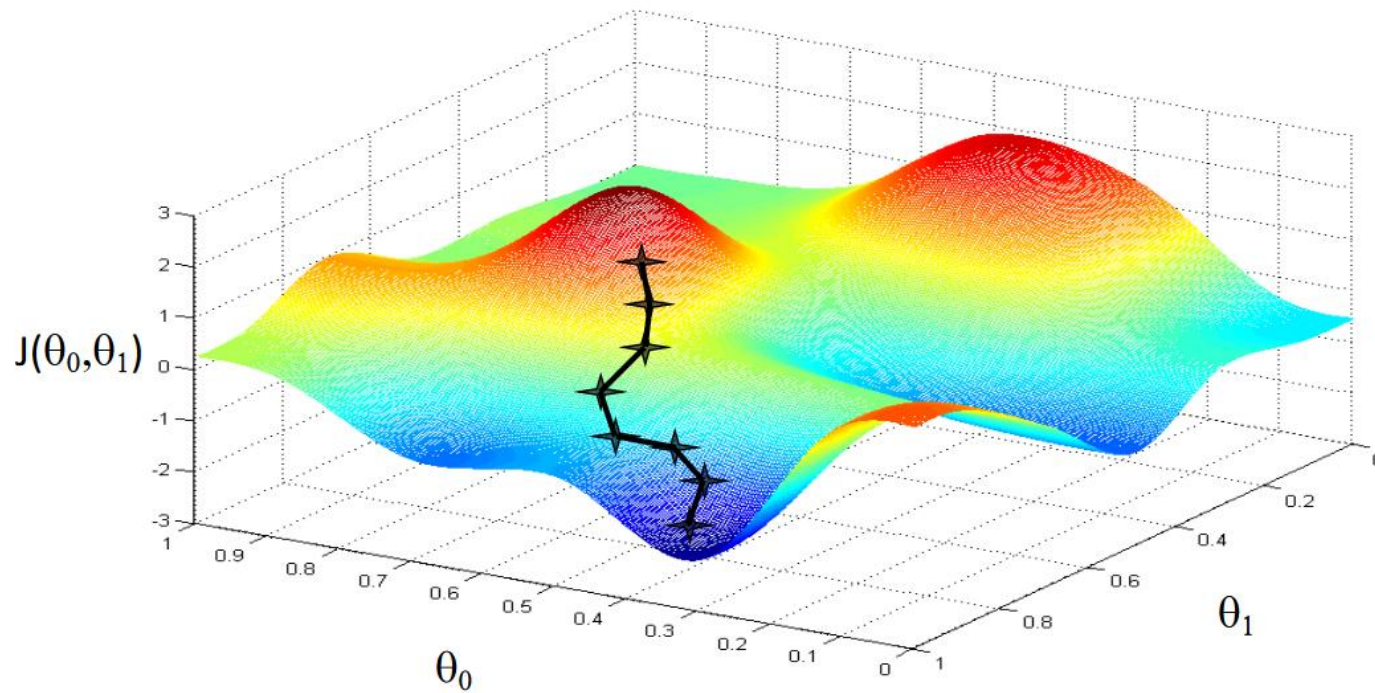


Gradient Descent for Linear Regression

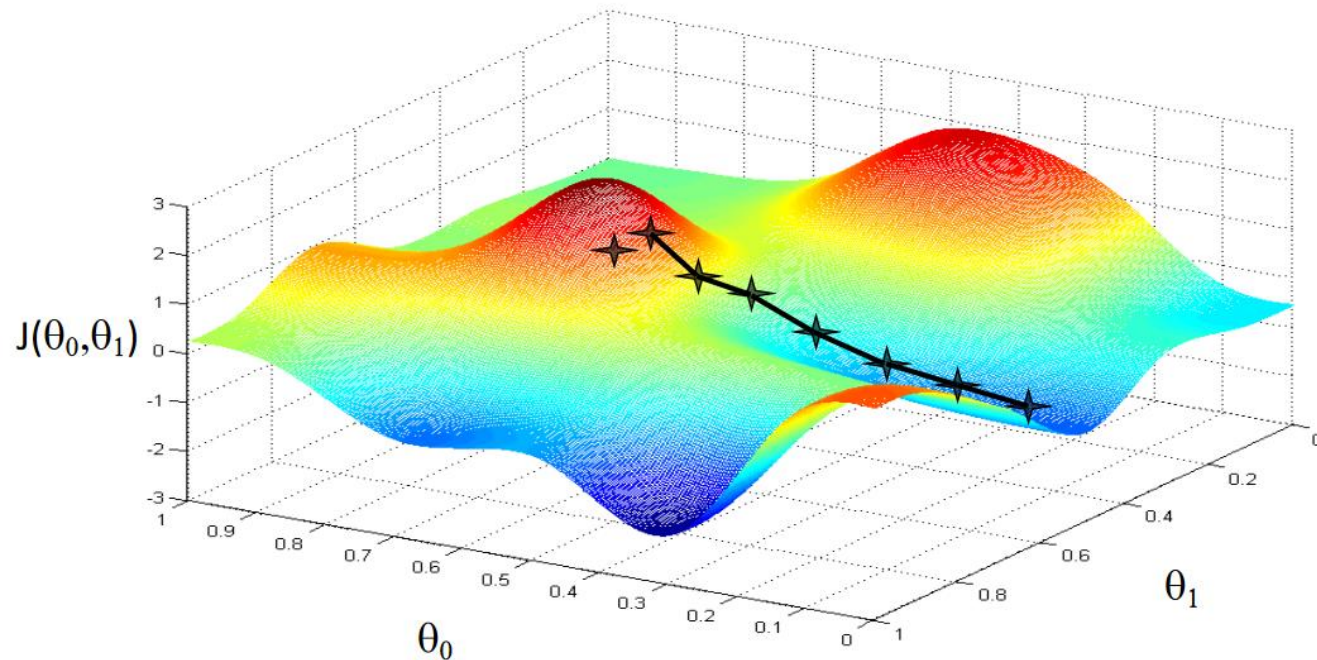
Reaching a minimum

- Choose initial value for θ
- Chose a new value for θ to reduce $J(\theta)$



Reaching a minimum

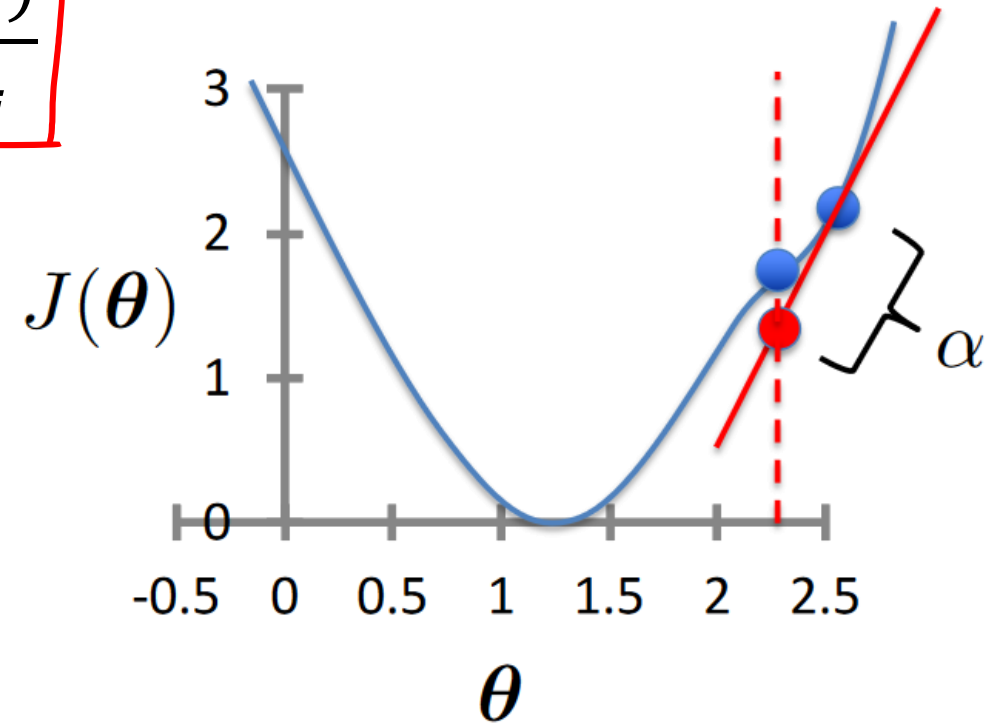
- Choose initial value for θ
- Chose a new value for θ to reduce $J(\theta)$



Gradient Descent

- Initialize θ
- Repeat until convergence

$$\theta_j := \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}$$



1
2
3

Derivatives: - $f(a) = 3a$



$$\begin{array}{l} a=2 \rightarrow f(a)=6 \\ a=2.001 \rightarrow f(a)=6.003 \end{array} \quad \left| \quad f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \right.$$

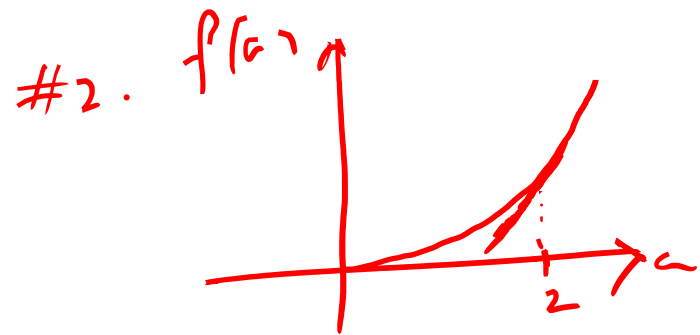
$$\text{Slope} \Big|_{a=2} = \frac{\text{height}}{\text{width}} = \frac{0.003}{0.001} = 3$$

$$a=5 \rightarrow f(a)=15$$

$$a'=5.001 \rightarrow f(a')=15.003$$

$$\text{Slope} \Big|_{a=5} = \frac{0.003}{0.001} = 3$$

$$\boxed{f'(a) = 3}$$



$$\boxed{f(a) = a^2} \longrightarrow \boxed{f'(a) = 2a}$$

$$a = 2 \longrightarrow f(a) = 4$$

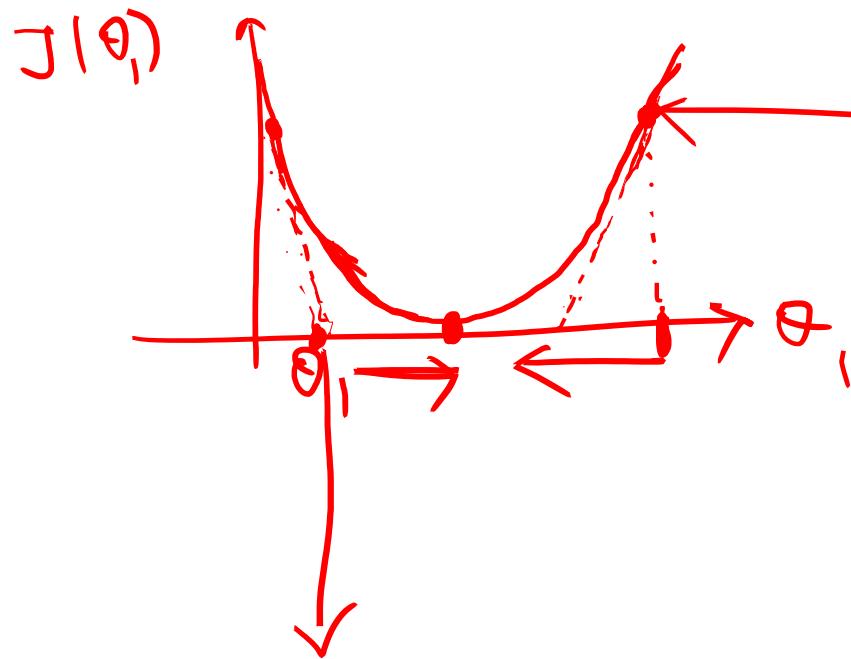
$$a = 2.001 \longrightarrow f(a) \approx 4.004$$

$$\left. \text{slope} \right|_{a=2} = \frac{0.004}{0.001} = 4$$

$$a = 5 \longrightarrow f(a) = 25$$

$$a = 5.001 \longrightarrow f(a) \approx 25.010$$

$$\left. \text{slope} \right|_{a=5} = \frac{0.010}{0.001} = 10$$



$$\theta_1 := \theta_1 - \alpha \frac{\partial J(\theta_1)}{\partial \theta_1}$$

$$\theta_1 := \theta_1 - \alpha k \quad [k \text{ is true}]$$

$$\theta_1 := \theta_1 - \alpha (-k) \quad [k \text{ is true}]$$

$$\theta_1 := \theta_1 + \alpha k$$

$$\left[\frac{\partial}{\partial \theta_j} J(\vec{\theta}) \right] = \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(h_{\theta}(\vec{x}^{(i)}) - y^{(i)} \right)^2$$

$$= \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^n \left(\sum_{k=0}^d \theta_k x_k^{(i)} - y^{(i)} \right)^2$$

$$= \frac{1}{2n} \sum_{i=1}^n 2 \left[\left(\sum_{k=0}^d \theta_k x_k^{(i)} - y^{(i)} \right) \times \frac{\partial}{\partial \theta_j} \left(\sum_{k=0}^d \theta_k x_k^{(i)} - y^{(i)} \right) \right]$$

$$= \frac{1}{n} \sum_{i=1}^n \left(\sum_{k=0}^d \theta_k x_k^{(i)} - y^{(i)} \right) x_j^{(i)}$$

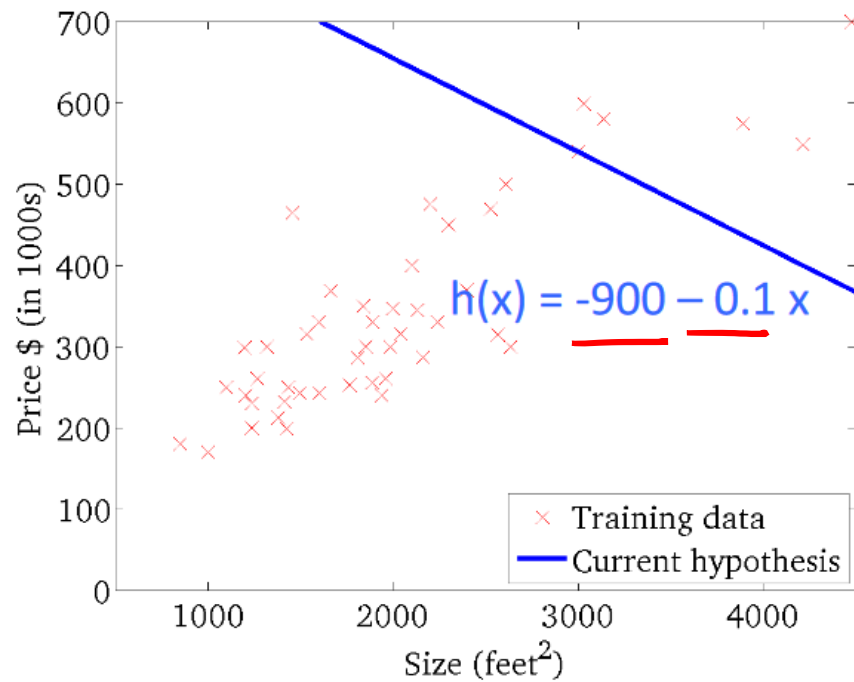
$$\theta_j := \theta_j - \alpha \frac{1}{n} \sum_{i=1}^n \left(\sum_{k=0}^d \theta_k x_k^{(i)} - y^{(i)} \right) x_j^{(i)}$$

$$\begin{aligned} & a_1 f(x_1) + a_2 f(x_2) \\ &= \underline{f(a_1 x_1 + a_2 x_2)} \\ & \quad \downarrow \\ & f \rightarrow \text{linear.} \end{aligned}$$

Gradient descent convergence

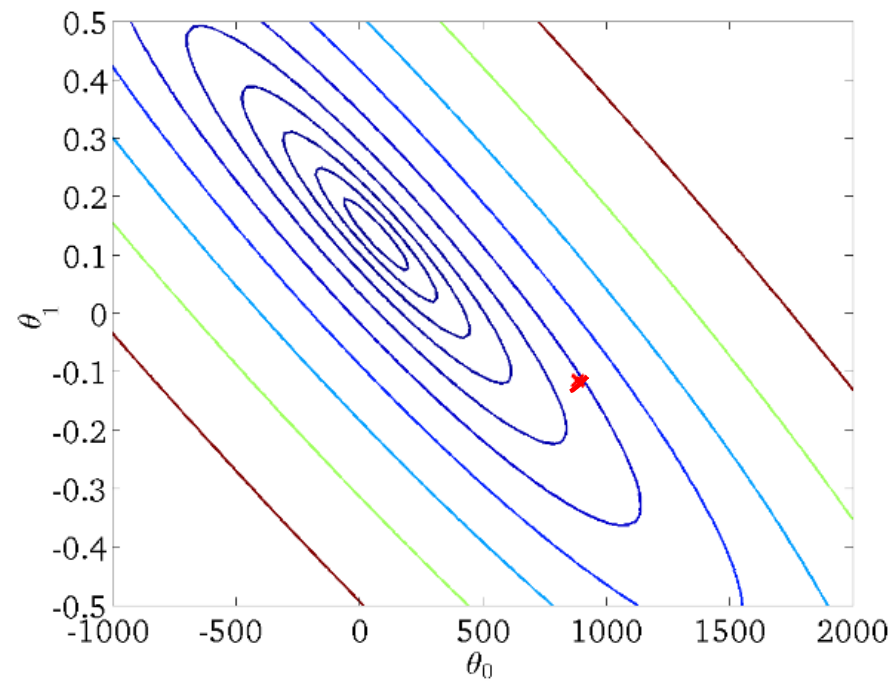
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

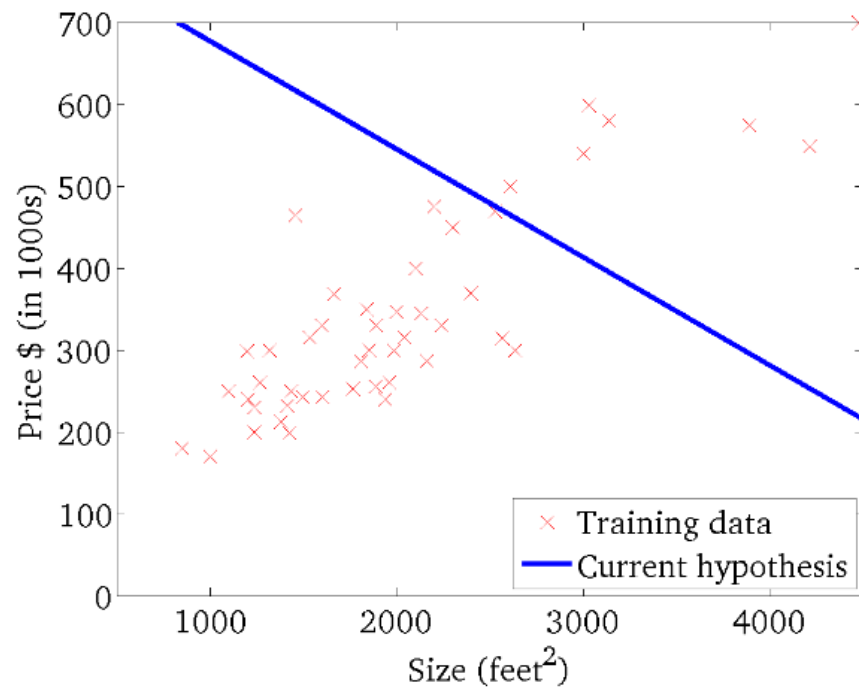
(function of the parameters θ_0, θ_1)



Gradient descent convergence

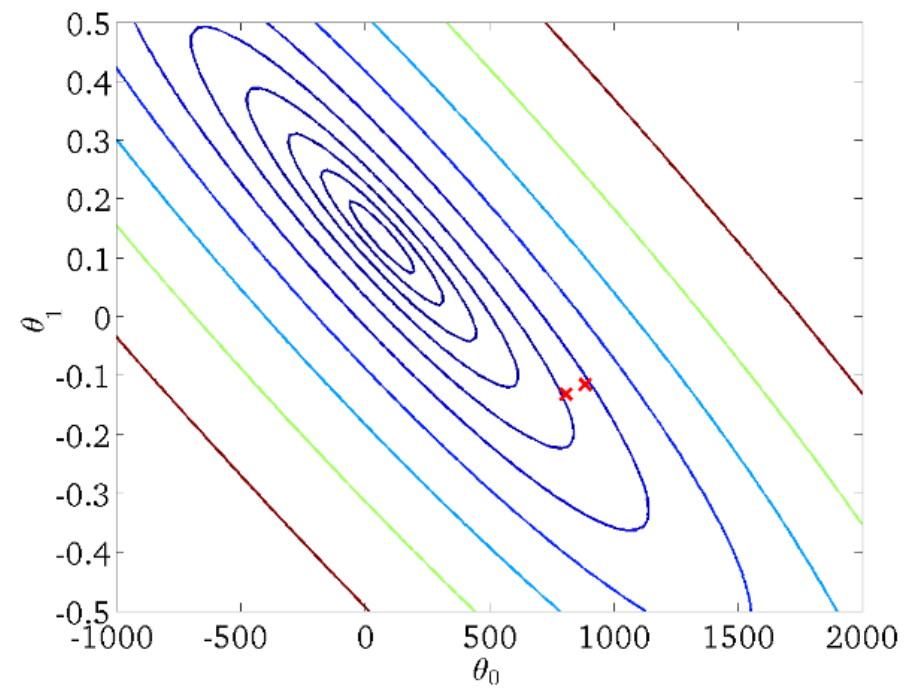
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

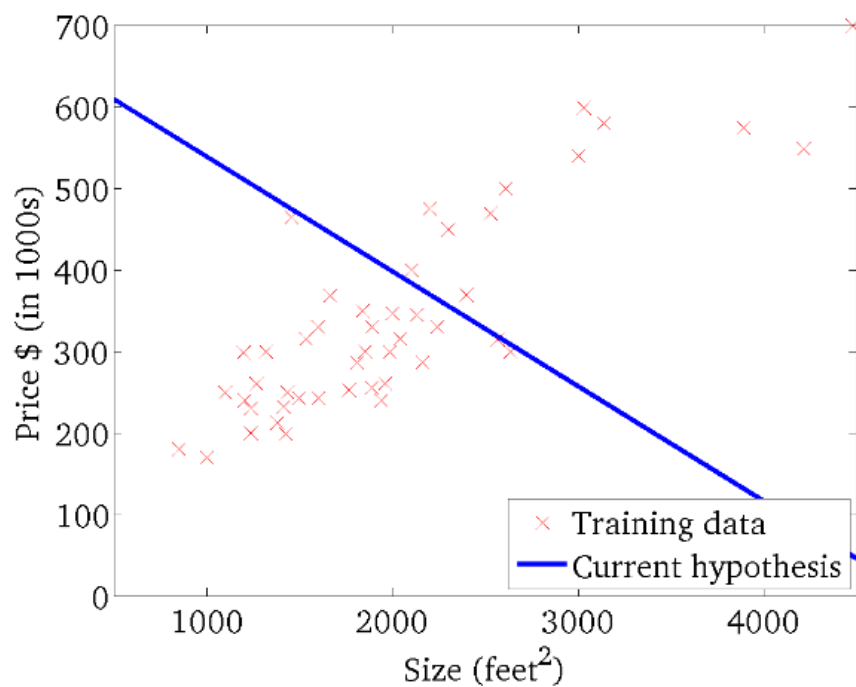
(function of the parameters θ_0, θ_1)



Gradient descent convergence

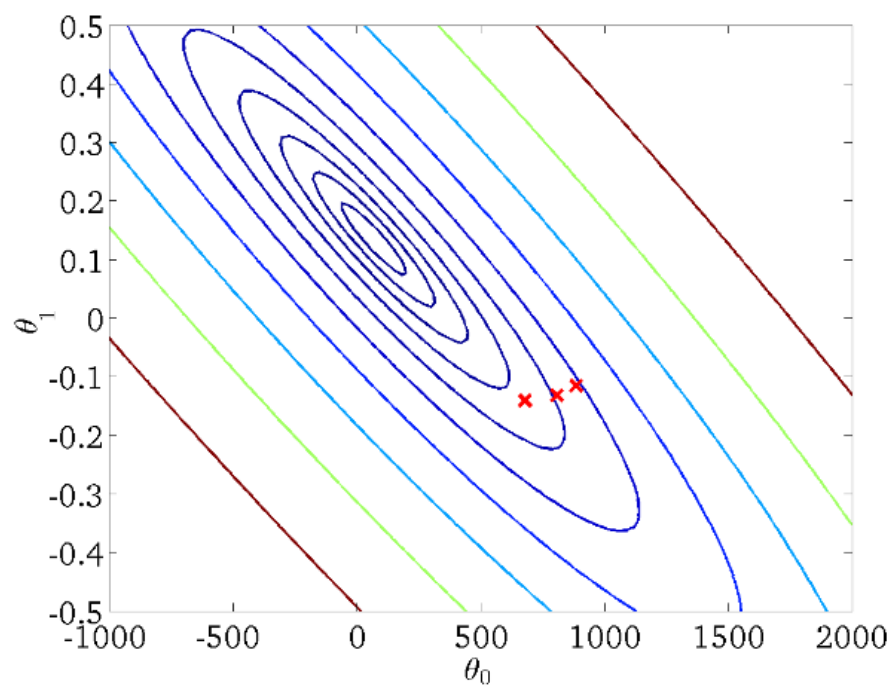
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

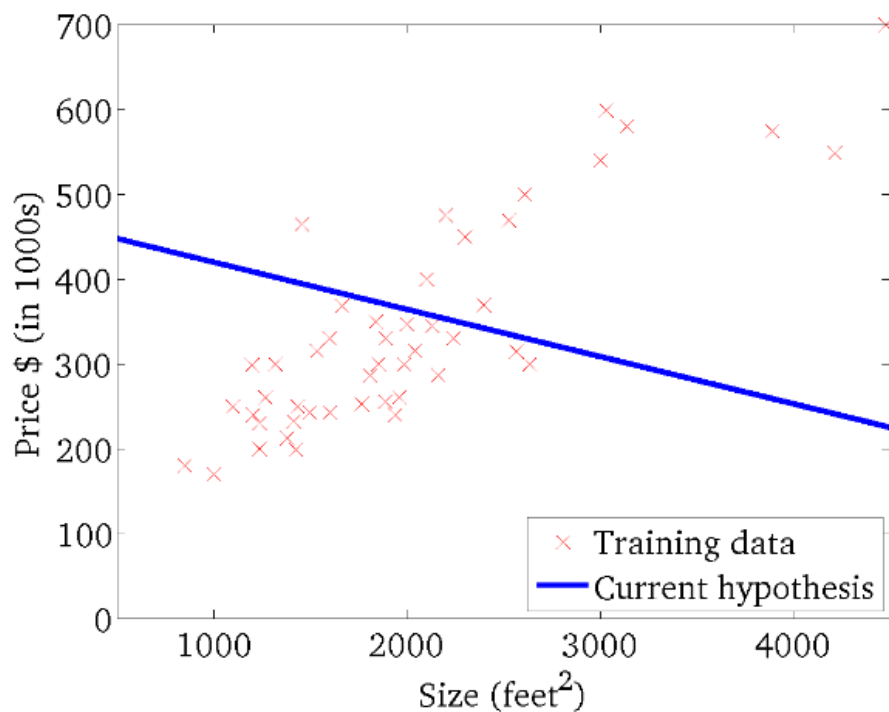
(function of the parameters θ_0, θ_1)



Gradient descent convergence

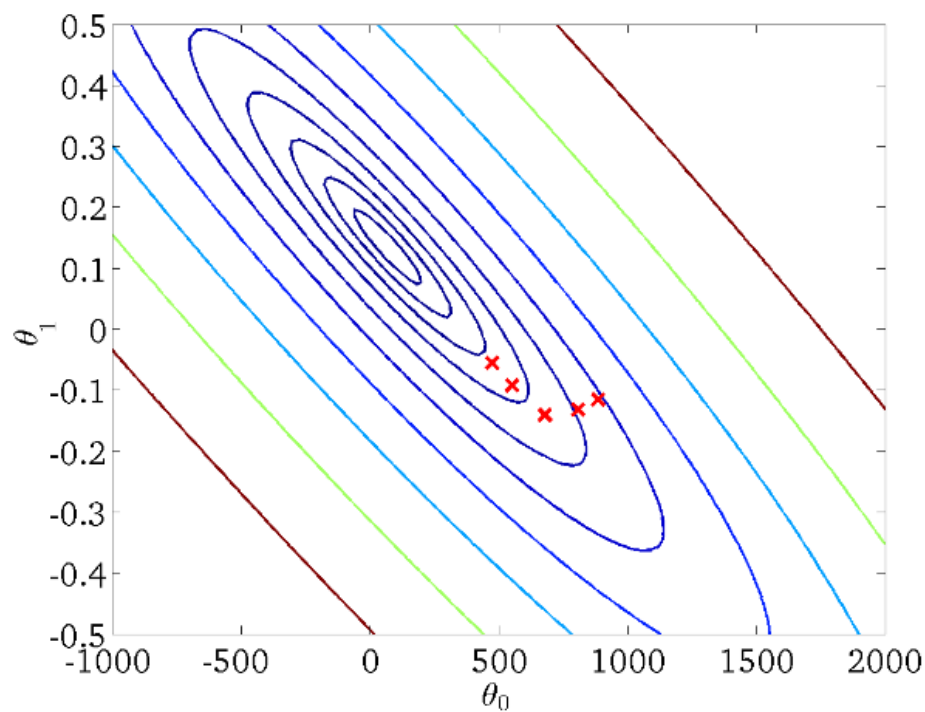
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)

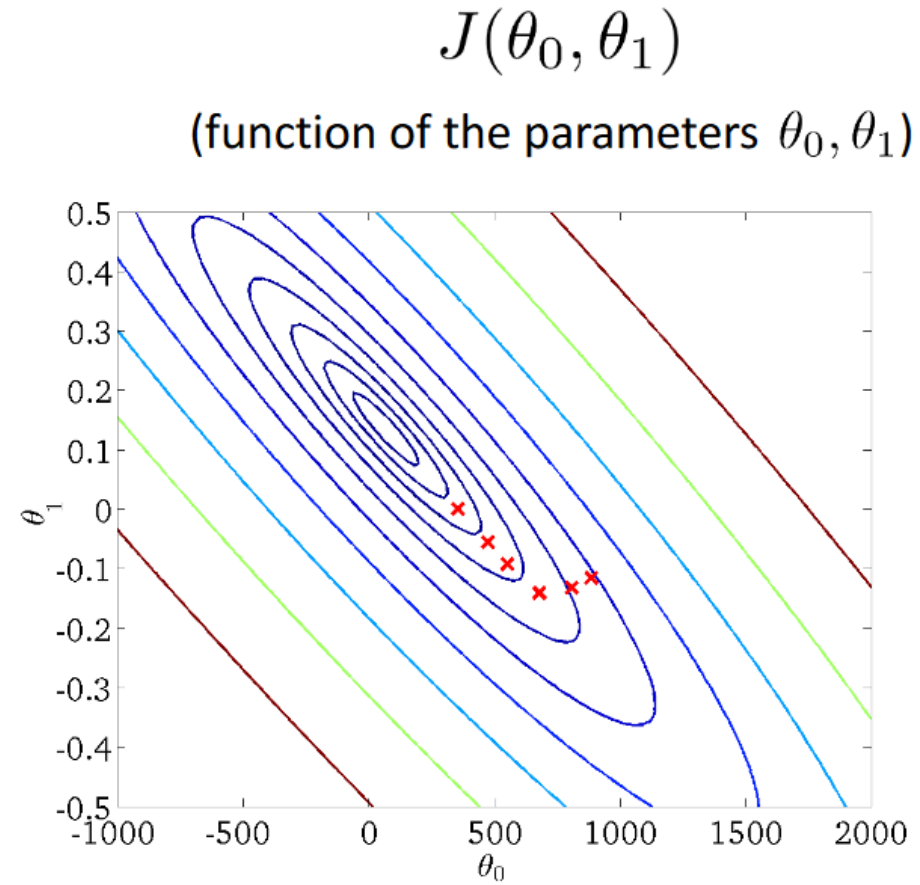
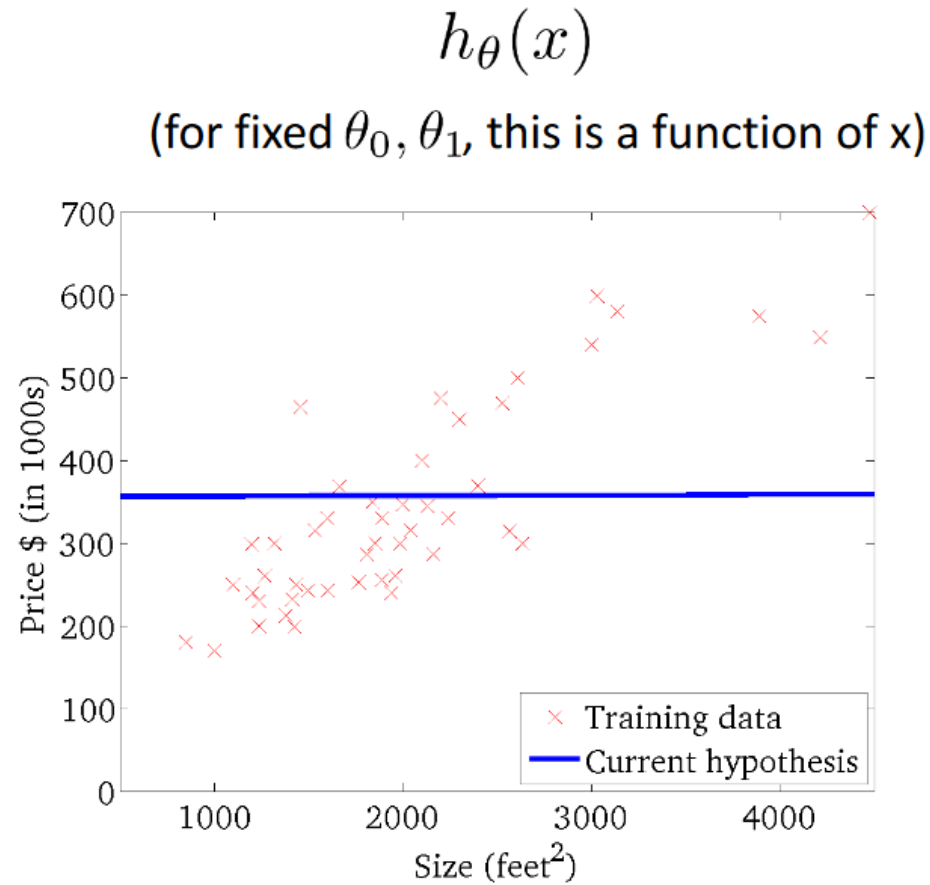


$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



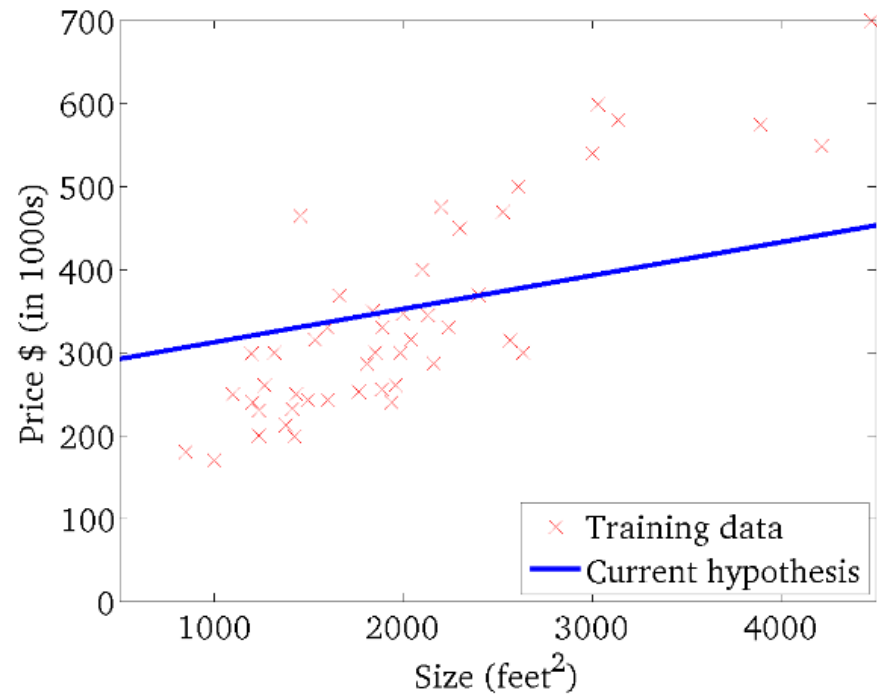
Gradient descent convergence



Gradient descent convergence

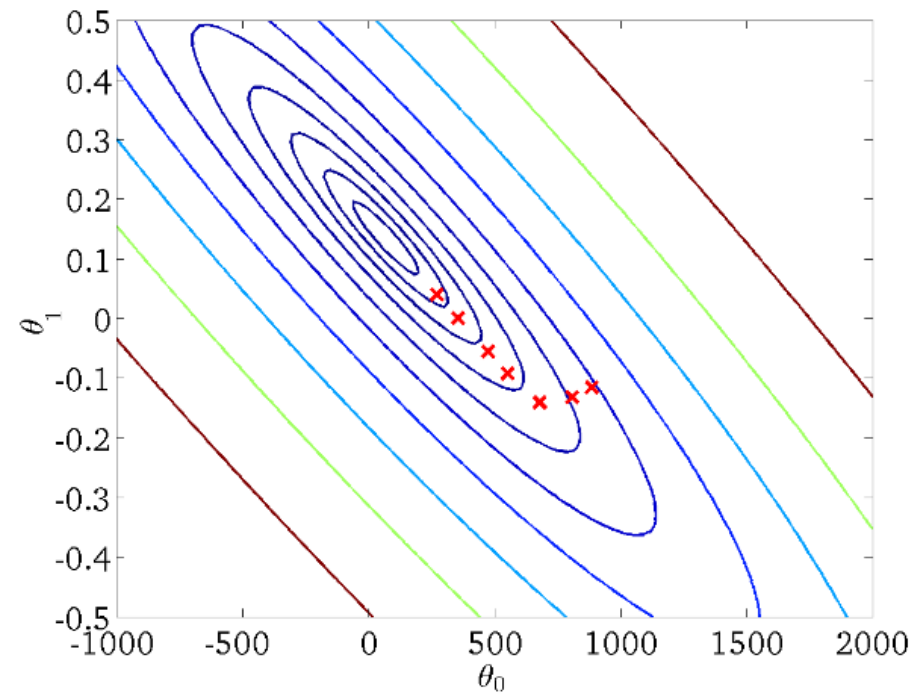
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

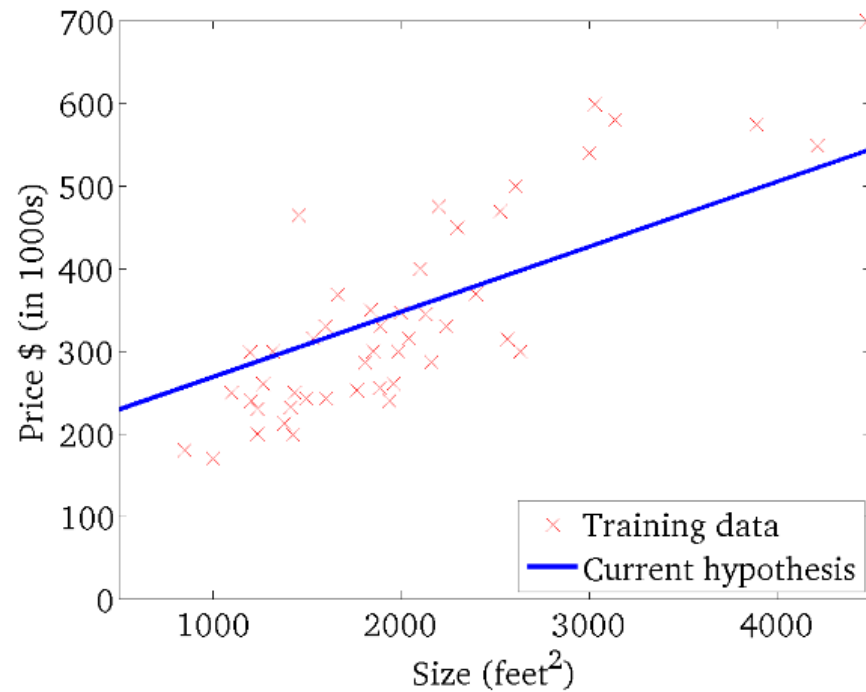
(function of the parameters θ_0, θ_1)



Gradient descent convergence

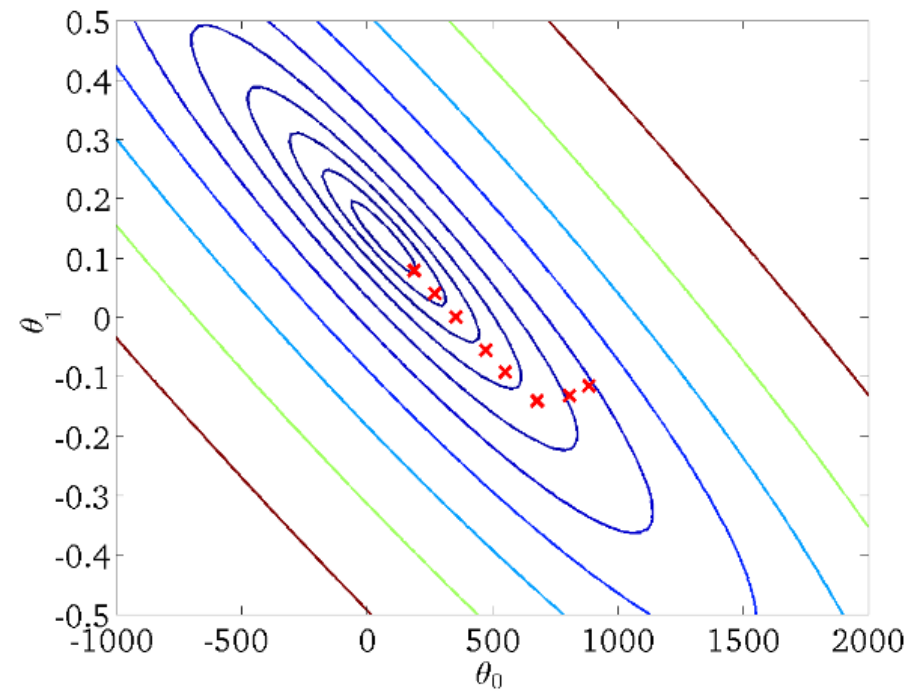
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

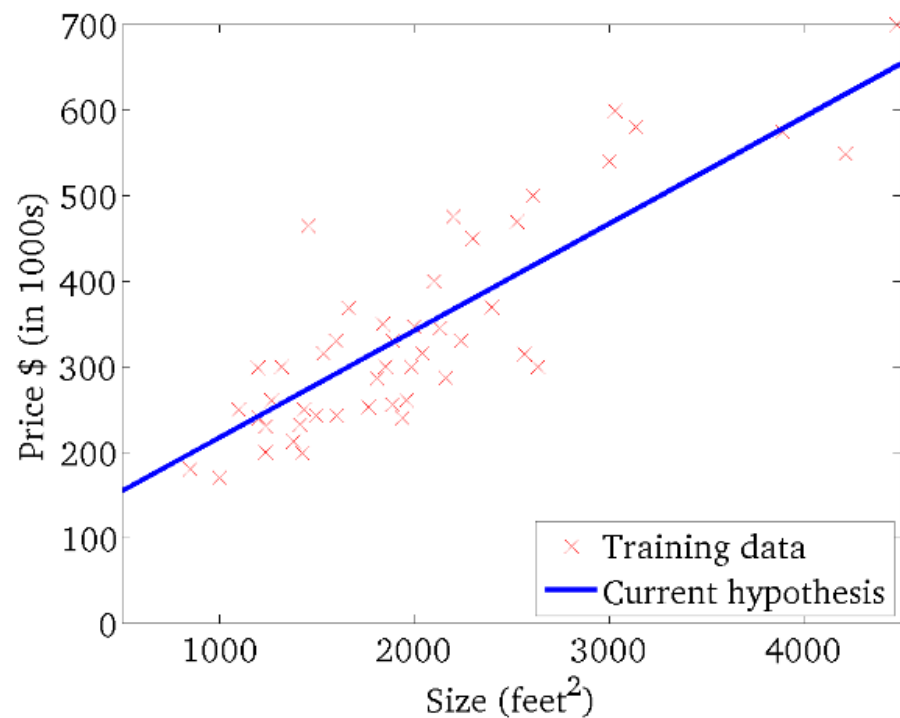
(function of the parameters θ_0, θ_1)



Gradient descent convergence

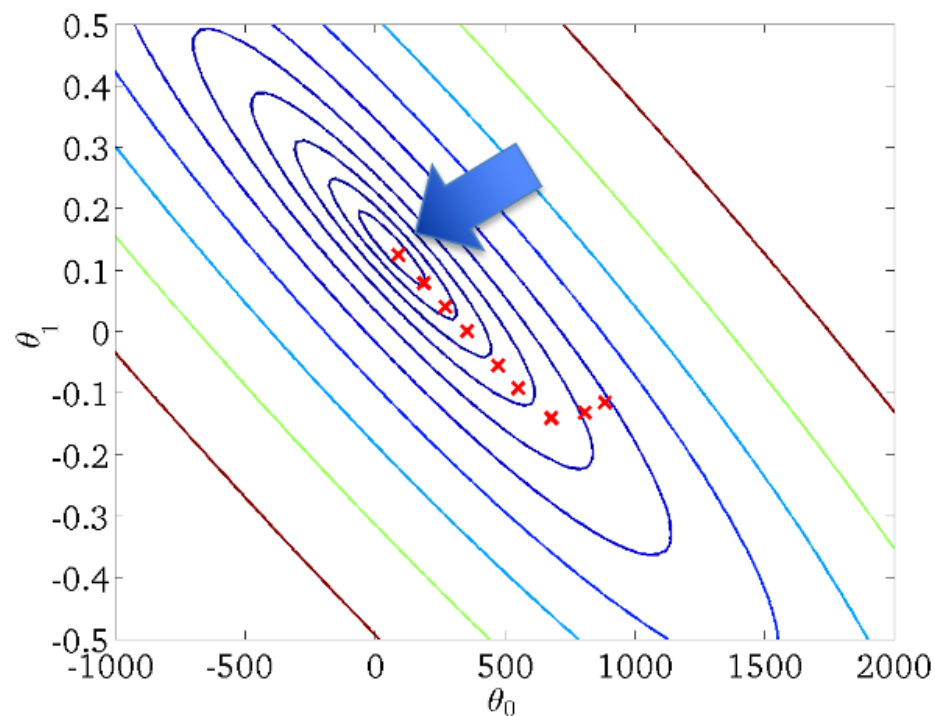
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



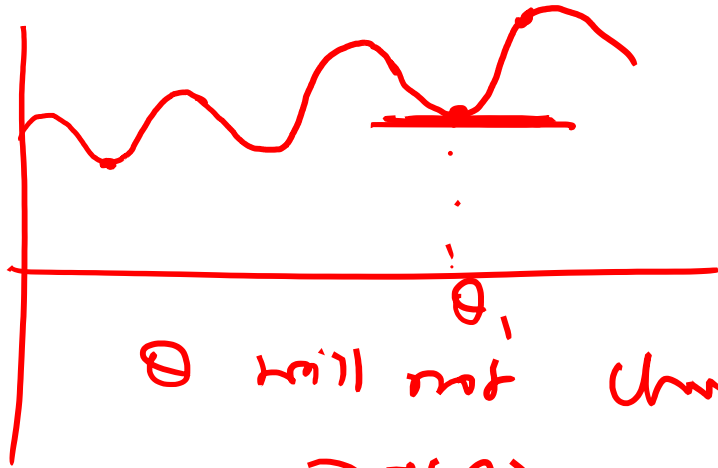
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)

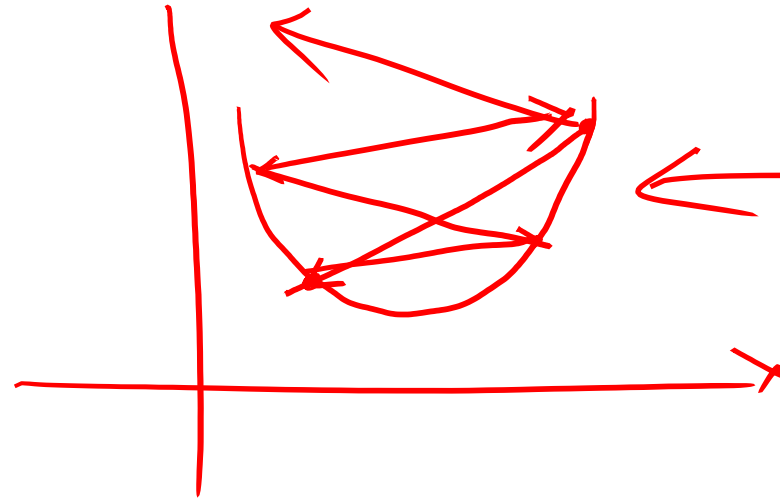


The term α :-

$$\theta_j := \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}$$



θ will not change
as $\frac{\partial J(\theta)}{\partial \theta_j} = 0$ at
 θ_1 .



Larger α
You may not
be able
converge
global
minima.